

Drivers create one file for each piece of device information that is to be exposed to the user space through `sysfs`. Drivers use a special API for file creation in the user space, and this is as follows:

```
int device_create_file(struct device *dev, const struct
device_attribute *attr);
```

...where,

dev: This is the pointer to the device structure of the driver module.

attr: This is the attribute which is to be created.

For removal of the file from the `sysfs` directory, the following interface can be used:

```
void device_remove_file (struct device *dev, const struct
device_attribute *attr)
```

Sysfs object relationship

Let's suppose there is the directory `/sys/kernel/example1`, the kobject pointer of which is `ex1_kobj` and there is a need to create a link file in the `/sys/kernel/example2` directory named 'ex2', which has to show the relationship with the `example1` directory. This can be accomplished by symbolic links in `sysfs`. Also, the kobject pointer of the `example2` directory is `ex2_kobj`.

Symbolic links are created in `sysfs` through the following interface:

```
int sysfs_create_link(struct kobject *kobj, struct kobject
*target, char *name);
```

...where,

kobj: This is the pointer to the kobject that represents the directory in which the link is to be created.

target: This is the target directory to which the link is to be pointed.

name: This is the name with which the link is to be created.

For our example, the link is created as follows:

```
sysfs_create_link(ex1_kobj, ex2_kobj, "ex2");
```

To delete symbolic links, the following interface is available:

```
void sysfs_remove_link(struct kobject *kobj, char *name);
```

In our example, to delete the link, the following function is called:

```
sysfs_remove_link(ex2_kobj, "ex2");
```



References

- [1] <https://www.kernel.org/doc/Documentation/filesystems/sysfs.txt>
- [2] <http://lx.missinglinkelectronics.com/linux/drivdr/rto/rto-ds1343.c>
- [3] <https://www.kernel.org/pub/linux/kernel/pepepe/mochel/doc/papers/ols-2005/mochel.pdf>

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